Athena Autocallable Notes on the CAC40 under a Calibrated Heston Model: Fair Coupons, Scenarios and Greeks

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November 2026

Abstract

We price single-underlying Athena autocallable notes, whose coupon is paid at the call only, on the CAC40 of 12 February 2025. The underlying follows a Heston model calibrated to the implied volatility surface (RMSE 0.42 volatility point), with a drift read from a forward curve bootstrapped from put–call parity. The note is described by a single parametrisation covering the observation frequency, snowball or fixed coupons, autocall barrier and step-down, non-call period, European, American continuous or American daily capital protection, airbag and capital guarantee. The notes are priced by Monte Carlo with antithetic variates and a control variate; the American protection is handled with the Brownian bridge crossing probability as a conditional expectation. The pricer recovers three closed-form limit cases within one standard error. We report the fair coupon (price at par) of 16 variants together with their scenario statistics: the base Athena (4 years, annual observations, autocall at 100%, European protection at 60%) pays 9.41% a year, is called at the first date with probability 55.7% and loses capital with probability 9.1%. Switching to an American protection adds 1.77 points of coupon. The note is short volatility (-0.55 per vol point on a nominal of 100) and short dividends (-1.12 per point of carry).

Contents

1 Introduction

In an AutoCallable note, the investor receives coupons well above the risk-free rate, and the note is redeemed early ("called") as soon as the index is above a reference level on an observation date. The coupon is financed by selling a down-and-in put. If the note is never called and the index ends below a protection barrier, the investor bears the loss of the index.

This report studies the **Athena**, whose coupon is paid only when the note is called, usually accumulated over time (*snowball*). The other main family, the Phoenix, whose coupon is paid at each observation date where the index is above a coupon barrier, is treated in a companion report [**?**]. On top of this structure, issuers combine many features: observation frequency, step-down of the autocall barrier, non-call period, European or American protection, airbag, capital guarantee. We will have a look on how to price these different versions in a single framework, on real market data, and measure how each feature moves the fair coupon, the probabilities of the scenarios and the Greeks.

The model is the Heston model calibrated in [**?**] to the CAC40 implied volatility surface, and the Monte Carlo engine is the one used for barrier options in [**?**]. The scripts and notebooks are open source; Section **??** describes the code.

2 Products

2.1 Common structure

A note of nominal N on an index S has an initial fixing S_{ref} and observation dates $0 < t_1 < \cdots < t_n = T$. The performance at t_k is $P_k = S_{t_k}/S_{\text{ref}}$. All barriers are fractions of S_{ref}: autocall barriers B_k^{call} and protection barrier B^{prot}. The dates t_k with $t_k \geq t_{\text{nc}}$ are callable, t_{nc} being the end of the non-call period; the maturity is always callable. The first callable date where $P_k \geq B_k^{\text{call}}$ is the call date τ.

Redemption at maturity. If the note is never called, it is redeemed at T at

$$R_T = N\left[(1 - \mathbf{1}_{\text{KI}}) + \mathbf{1}_{\text{KI}} \max\Big(\phi,\ \min\left(1,\ P_T/K_{\text{put}}\right)\Big)\right], \tag{1}$$

where $\mathbf{1}_{\text{KI}}$ is the knock-in event of the protection, K_{put} the strike of the short put and ϕ the capital floor. The knock-in event is

- *European protection*: $P_T < B^{\text{prot}}$;
- *American protection*: the index touches $B^{\text{prot}} S_{\text{ref}}$ at any time before maturity, continuously or on daily closes.

The standard note has $K_{\text{put}} = 1$ and $\phi = 0$: below the barrier, the investor receives NP_T and the redemption jumps from N to NB^{prot} at the barrier. The *airbag* note has $K_{\text{put}} = B^{\text{prot}}$: the redemption is continuous at the barrier and equal to NP_T/B^{prot} below it. A *capital guarantee* sets $\phi > 0$. Figure **??** draws the three profiles.

2.2 Athena

The Athena pays a coupon only with the redemption at the call date. If $\tau = t_k$, the investor receives

$$N(1 + c_k), \qquad c_k = \begin{cases} c\, t_k & \text{snowball (the coupons accumulate),} \\ c\,(t_k - t_{k-1}) & \text{fixed coupon,} \end{cases} \tag{2}$$

with c the coupon rate per annum. If the note is not called, no coupon is paid and the redemption is (**??**). A call at maturity ($P_T \geq B_n^{\text{call}}$) pays the last coupon.

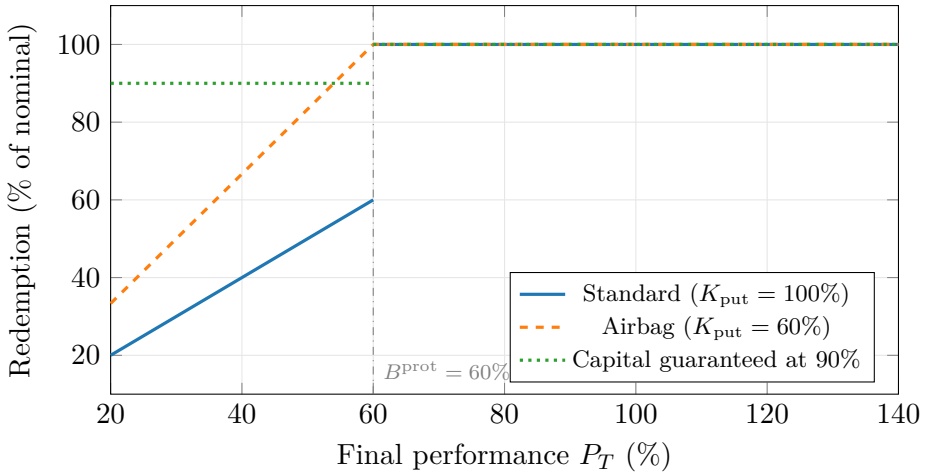


Figure 1: Redemption at maturity of a note that was never called, European protection at 60%.

2.3 Parametrisation

Table **??** lists the parameters of the class `Athena`; every variant of Section **??** is one of these parameters changed.

Feature	**Parameter**	**Values**
Maturity, initial fixing, nominal	`maturity`, `initial_fixing`, `nominal`	years, index level, 100
Observation dates	`frequency` or `observation_dates`	monthly to annual, or any list ending at T
Coupon rate	`coupon_rate`	per annum, fraction of the nominal
Athena coupon	`coupon_type`	`snowball`, `fixed`
Autocall barrier	`autocall_barrier`, `step_down`	fraction of S_{ref}, decrease per observation
Non-call period	`first_call_date`	years
Capital protection	`protection_barrier`, `protection_type`	fraction of S_{ref}; `european`, `american`, `american_discrete`
Airbag, guarantee	`put_strike`, `floor`	fraction of S_{ref}, fraction of the nominal

Table 1: Parameters of the Athena (`Products/Athena/athena.py`).

3 Market Data and Model

The data are the CAC40 option chain of 12 February 2025 (spot $S_0 = 8042.19$, 142 quotes on 13 expiries from 9 days to 4.86 years) and the EURIBOR 6M zero-coupon curve. The forward curve $F(0,t)$ is bootstrapped from put–call parity with a piecewise constant borrow-plus-dividend yield $q(t)$ [**?**]; its integral reaches 0.136 at 4.86 years, so dividends lower the 4-year forward by about 11% with respect to a zero carry. Under $\mathbb{Q}$,

$$\frac{\mathrm{d}S_t}{S_t} = \partial_t \ln F(0,t)\,\mathrm{d}t + \sqrt{v_t}\,\mathrm{d}W_t^S, \qquad \mathrm{d}v_t = \kappa(\theta - v_t)\,\mathrm{d}t + \xi\sqrt{v_t}\,\mathrm{d}W_t^v, \qquad \mathrm{d}\langle W^S, W^v\rangle_t = \rho\,\mathrm{d}t, \tag{3}$$

with the parameters of Table **??**. The notes are struck at the money: $S_{\mathrm{ref}} = S_0 = 8042.19$, and S_{ref} stays fixed when the Greeks bump the market. All the dates of the 4-year notes lie before the last listed expiry (4.86 years).

v_0	θ	κ	ξ	ρ	IV RMSE
0.0197	0.0438	1.308	0.482	-0.605	0.421 vol pts

Table 2: Heston parameters calibrated on the CAC40 surface of 12 February 2025 [**?**].

4 Pricing Method

4.1 Simulation

The paths of (S, v) are simulated with an Euler full truncation scheme on a weekly grid (52 steps a year, 208 steps for 4 years), the drift over each step being $\ln(F(0,t_{i+1})/F(0,t_i))$, so that $\mathbb{E}[S_t] = F(0,t)$ at every date. $N = 50{,}000$ paths with antithetic variates are used (seed 42); prices and standard errors are computed on the 25,000 independent pair averages. The observation dates are rounded to the grid; with weekly steps, every monthly, quarterly, semi-annual and annual date falls within half a week of its contractual date.

4.2 Cash flows and protection

For each path, the note is followed through its observation dates, each cash flow being discounted with $D(0,t_k) = e^{-r(t_k)t_k}$. The knock-in indicator of (**??**) is

- *European*: $\mathbf{1}_{\{P_T < B^{\mathrm{prot}}\}}$;
- *American continuous*: replaced by its conditional probability given the simulated path, $1 - \prod_i (1 - p_i)$, where $p_i = \exp(-2\ln(S_{t_i}/H)\ln(S_{t_{i+1}}/H)/(v_{t_i}\Delta t))$ is the probability that the Brownian bridge of $\ln S$ touches $H = B^{\mathrm{prot}} S_{\mathrm{ref}}$ between two grid dates [**?**];
- *American daily*: the barrier is checked on a daily grid (252 steps a year).

4.3 Variance reduction

Besides antithetic variates, the discounted European put struck at $B^{\mathrm{prot}} S_{\mathrm{ref}}$, whose Heston price is known semi-analytically (Lewis formula), is used as a control variate with an optimal coefficient estimated on the paths.

4.4 Fair coupon and scenario statistics

On a fixed set of paths, the call dates do not depend on the coupon, and every cash flow is affine in the coupon rate c: the price is $V(c) = V(0) + c\,(V(1) - V(0))$, and the coupon that prices the note at par is

$$c^\star = \frac{N - V(0)}{V(1) - V(0)}, \tag{4}$$

obtained with two valuations. It excludes the issuer margin and funding spread. The same paths give the probability of a call at each date, the probability to reach maturity, the probability of a capital loss ($\mathbf{1}_{\mathrm{KI}} = 1$ and $P_T < K_{\mathrm{put}}$), the expected loss at maturity, the expected life $\mathbb{E}[\min(\tau, T)]$ and the expected coupon paid.

4.5 Greeks

The Greeks are central finite differences with common random numbers [**?**], on a nominal of 100: Delta and Gamma (spot $\pm 1\%$, carry kept), Vega and Volga ($\sqrt{v_0}$ and $\sqrt{\theta}$ shifted by ± 1 vol point), Vanna, Rho ($\pm$10bp on the zero curve, per $+1\%$), Dividend ($\pm$10bp on $q(t)$, per $+1\%$) and Theta (dates moved by one day). Delta is given per index point: multiply by $S_0/100 = 80.4$ for the value change of a 1% move. Each Greek is reported with its Monte Carlo standard error.

4.6 Validation

Three limit cases have a closed form (Table **??**). In a flat Black–Scholes market ($S_0 = S_{\mathrm{ref}} = 100$, $r = 3\%$, $q = 1\%$, $\sigma = 20\%$, $T = 2$, $N = 100$, Heston with $v_0 = \theta = \sigma^2$ and $\xi = 10^{-4}$, 10^5 paths):

- a note never called (autocall barrier out of reach) with a European protection at $B = 60$ is worth $ND(T) - \frac{N}{S_{\text{ref}}}\left[\text{Put}(B) + (S_{\text{ref}} - B)\,\text{DigitalPut}(B)\right]$;
- with an American protection, $ND(T) - \frac{N}{S_{\text{ref}}}\,\text{PDI}(K = S_{\text{ref}}, H = B)$ (Reiner–Rubinstein);
- a note called for sure at t_1 is worth $N(1 + c\,t_1)D(t_1)$.

Case	**Reference**	**Monte Carlo**	**Std error**	**Error / std error**
Athena never called, European protection (flat)	92.6369	92.6480	0.0141	0.79
Athena never called, American protection (flat)	91.5500	91.5483	0.0220	−0.07
Athena called for sure at $t_1 = 0.5$, $c = 8\%$ (flat)	102.4516	102.4516	0	exact

Table 3: Validation of the pricer.

5 Athena : Base case

Let us consider, under the risk-neutral measure $\mathbb{Q}$, the general case of an Athena in the base case. To lighten notation, we denote by

$$E_i = \bigcap_{j=1}^{i-1} \{X_j < B_{ac}\}, \qquad i = 1, \dots, N, \qquad E_1 = \Omega, \tag{5}$$

the event that the product has not been called before date t_i. The event $\{\tau = t_i\}$ of an early redemption at date $t_i < T$ is then $E_i \cap \{X_i \geq B_{ac}\}$, and $\{\tau = T\} = E_N$.

5.1 Price of the Athena

For a coupon c, the price of the Athena at inception is

$$P_0 = \mathbb{E}^{\mathbb{Q}}\big[D(\tau)\,\Pi_\tau\big], \tag{6}$$

which, writing the autocall events explicitly, expands as

$$\begin{aligned} P_0 = \sum_{i=1}^{N-1} & D(t_i)\,(1+ic)\,\mathbb{Q}(E_i \cap \{X_i \geq B_{ac}\}) \\ & + D(T)\,(1+Nc)\,\mathbb{Q}(E_N \cap \{X_N \geq B_{ac}\}) \\ & + D(T)\,\mathbb{Q}(E_N \cap \{B_p \leq X_N < B_{ac}\}) \\ & + D(T)\,\mathbb{E}^{\mathbb{Q}}\big[X_N\,\mathbf{1}_{E_N \cap \{X_N < B_p\}}\big]. \end{aligned} \tag{7}$$

The first term corresponds to an early redemption at date t_i, the second to a redemption with coupons at maturity, the third to the repayment of the notional only, and the last one to the capital loss scenario.

5.2 Linearity in the coupon and embedded put

We now turn to the determination of the fair coupon level c. A basic Athena has the convenient property that its price is affine in the coupon. Indeed, the coupon only affects the *amount* paid at redemption, never the redemption time τ, which depends on the path of the underlying alone. Moreover, on the event E_N that the product has not been called before maturity, the terminal payoff can be rewritten, since $B_p < 1$, as

$$\Pi_T = 1 + Nc\,\mathbf{1}_{\{X_N \geq B_{ac}\}} - (1 - X_N)^+\,\mathbf{1}_{\{X_N < B_p\}}. \tag{8}$$

The last term is the payoff of a down-and-in put with strike 100% and barrier B_p observed at maturity, which is sold by the investor. Substituting (**??**) into (**??**) yields the decomposition

$$P_0 = \underbrace{\mathbb{E}^{\mathbb{Q}}\big[D(\tau)\big]}_{\text{notional}} + c \underbrace{\sum_{i=1}^{N} i\,D(t_i)\,\mathbb{Q}(E_i \cap \{X_i \geq B_{ac}\})}_{B\ \text{(coupons)}} - \underbrace{D(T)\,\mathbb{E}^{\mathbb{Q}}\Big[(1 - X_N)^+\,\mathbf{1}_{\{X_N < B_p\}}\,\mathbf{1}_{E_N}\Big]}_{\text{PDI}^{ac}}. \tag{9}$$

The investor is thus long a note repaying the notional at the random time τ, long a strip of conditional coupons, and short a down-and-in put PDI^{ac}. Grouping the terms that do not depend on the coupon, the price takes the affine form

$$P_0 = A + c\,B, \qquad A = \mathbb{E}^{\mathbb{Q}}\big[D(\tau)\big] - \text{PDI}^{ac}, \tag{10}$$

where A is the price of the same product with a zero coupon, and B the value of a unit coupon paid in all scenarios where coupons are received.

5.3 Fair coupon

The fair coupon $c^\star$ is the coupon for which the price equals a target $P^\star$, namely $P^\star = 1$ for a fair price, or $P^\star = 1 - m$ when the issuer retains a margin m. By linearity, it is given in closed form by

$$\boxed{c^\star = \frac{P^\star - A}{B} = \frac{P^\star - \mathbb{E}^{\mathbb{Q}}\big[D(\tau)\big] + \mathrm{PDI}^{ac}}{B}} \tag{11}$$

To determine the root of this equation, a single Monte Carlo simulation provides A and B, hence $c^\star$. This expression also makes the economics of the product transparent: the coupons finance the down-and-in put sold by the investor, together with the funding benefit of the issuer, since $\mathbb{E}^{\mathbb{Q}}[D(\tau)] < 1$ as soon as cash flows are discounted with a positive rate and funding spread.

5.4 Remarks on the embedded put

The put PDI^{ac} is not a vanilla barrier option: it is knocked out whenever the product is autocalled, which is captured by the indicator $\mathbf{1}_{E_N}$. Since $\mathbf{1}_{E_N} \leq 1$, its value is bounded by that of the standard European down-and-in put,

$$\mathrm{PDI}^{ac} \leq \mathrm{PDI} = D(T)\,\mathbb{E}^{\mathbb{Q}}\Big[(1 - X_N)^+\,\mathbf{1}_{\{X_N < B_p\}}\Big]. \tag{12}$$

The latter itself decomposes into a vanilla put struck at the barrier and a digital put,

$$(1 - X_N)^+\,\mathbf{1}_{\{X_N < B_p\}} = (B_p - X_N)^+ + (1 - B_p)\,\mathbf{1}_{\{X_N < B_p\}}, \tag{13}$$

so that, absent the autocall feature, its value follows directly from the implied volatility surface around strike B_p, the digital term making it sensitive to the skew.

5.5 Structuration :

From the investor's perspective, the basic Athena note is composed of:

- a long position in an *autocallable zero-coupon note* with maximal maturity T, which repays the notional together with the accrued coupons at the first observation date t_i such that $X_i \geq B_{ac}$, and the notional alone at maturity otherwise;
- a short position in a *down-and-in put* (PDI) with strike 100% and barrier B_p observed at maturity, which is knocked out whenever the product is autocalled.

5.6 Sensitivity analysis :

We will now look at the different sensitivities of the product and what are the impacts of a move on certain parameters on c. We are now thinking from the investor point of view and with an already contracted note.

Volatility

The investor is short PDI then he is short ν. The PDI price decreases with volatility so an increase of the volatility will decrease the value of the note.

If the product is not contracted yet it will incread the proba to call the product and then it increases the coupon paid to the investor.

Protection barrier

If the protection barrier is higher, then it is more probable to cross it and it reduces the price of the note. The effect on the coupon c is that it is increased.

Recall Barrier

If the recall barrier is increased, it is less probable to be recall before maturity, then it increases the coupon c.

Dividend

If the underlying asset increases its dividend, it increases the value of c

Peut etre parce que l'investisseur ne touche pas les dividendes donc demandera une rémunération plus élevée.

Spread

Risk free rate

Maturity

Skew

Correlation

5.7 Pricing : A numerical example

Let us now consider a synthetic Athena issued by a risky issuer, which requires taking into account the credit spread of the issuer. The product is written on the CAC 40 index (dividends not reinvested). Its initial level is fixed on 31 January 2025 and it matures in November 2027. A coupon c accrues every quarter and is paid upon redemption. The underlying is observed quarterly: at each observation date, the product is redeemed early if the index closes at or above 100% of its initial level. If the product has not been called, the investor receives at maturity the notional plus the accrued coupons if the index is at or above its initial level, the notional alone if it is at or above the protection barrier $B_p = 60\%$, and the notional reduced by the full decline of the index otherwise.

5.7.1 Pricing

For the purpose of the pricing of this product, we will use the script `athena.py`[1] to build the product and `monte_carlo.py`[2] to simulate the different paths.

In this first basic case, we have the following result:

We also plotted the distribution of call probability over the different paths.

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Figure 2: Distribution of the call probability

As we can see, the probability for the product to be called during the first observation period is around 55% and the cumulative probability of call is around 86%.

[1] https://github.com/benguedalauzonelias/portfolio/tree/main/Products/Athena
[2] https://github.com/benguedalauzonelias/portfolio/tree/main/Equity_Models

		Base case
Contract	Maturity (years)	4
	Number of observations	16
	Observation frequency (years)	0.25
	Coupon (% p.a.)	6
	Coupon type	Snowball
	First autocall barrier (%)	100
	Last autocall barrier (%)	100
	Step-down per observation (%)	0
	First call date (years)	0.25
	Protection barrier (%)	60
	Protection type	European
	Redemption if knocked-in	Standard
	Capital guaranteed (%)	0
	Buffer (%)	0
	Range accrual (%)	–
	Duo barrier (%)	–
	Duo coupon ratio	–
	Look-back	–
	Nominal	100
Pricing	Price (% nominal)	97.80
	95% confidence interval (±)	0.11
	Fair coupon (% p.a.)	10.54
Scenarios	Probability to reach maturity (%)	13.29
	Probability of a capital loss (%)	5.22
	Expected loss at maturity (% nominal)	3.05
	Expected life (years)	1.03
	Expected number of coupons	0.87
	Expected coupon paid (% nominal)	3.01

Table 4: Contract features, pricing results and scenario statistics of the base-case Athena.

We then present the distribution of the different paths:

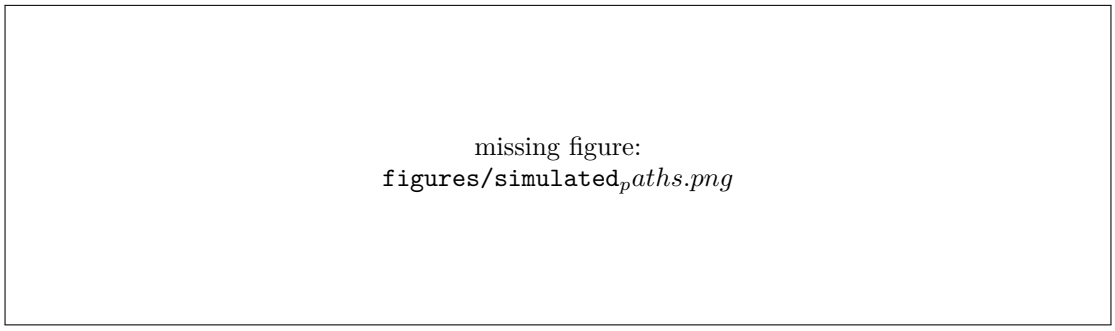


Figure 3: Distribution of the different paths

The expected life of this product is pretty short and because the investor is short a PDI, when the product losses money it losses a large amount of money.

5.7.2 Sensitivities

We are now interested in how the different parameters impact the price of a basic Athena product. We notice that the price after the emission of the note and the fair-coupon c^* move at the opposite one to another.

Protection barrier

We plotted the different parameters against a move of the protection barrier:

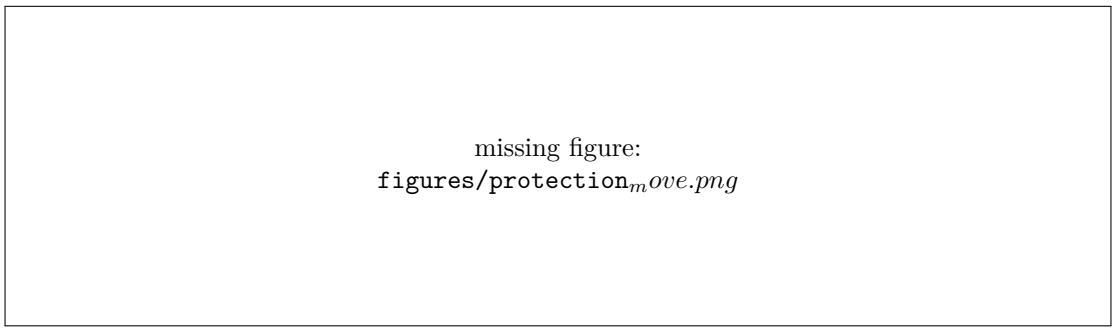


Figure 4: Move of the protection barrier

As we can see an augmentation of the protection barrier increases the value of c^*. It is easy to understand, in fact an increase of the protection barrier exposes more the investor to do not receive the coupon at the redemption date, the investor then takes more risk and must be remunerated for this risk. As well, the probability of capital loss and expected loss increasein the same direction as the barrier option.

Protection barrier (%) (% p.a.) capital loss (%) (% nominal)	Fair coupon Probability of a Expected loss		
10	4.99	0.10	0.09
20	5.55	0.48	0.41
30	6.38	1.09	0.87
40	7.58	2.07	1.50
50	9.00	3.43	2.25
55	9.81	4.32	2.67
60	**10.54**	**5.22**	**3.05**
65	11.24	6.19	3.42
70	11.94	7.32	3.78
75	12.48	8.34	4.07
80	13.00	9.57	4.34
90	13.64	11.81	4.68
100	13.80	13.29	4.77

Table 5: Fair coupon and loss statistics as a function of the protection barrier, all other parameters being those of the base case (in bold). The expected life of the product, equal to 1.03 years, does not depend on the protection barrier.

Autocall barrier

We also plotted the reaction of the different parameters for a move of the autocall barrier.



Figure 5: Move of the autocall barrier

5.7.3 Hedging

If we think from the desk's point of view there are long a PDI, their main sensibility to manage is the delta.

For real hedging of this kind of product, have a look to the paper pricing of barrier options [3]

[3] https://portfolio-eliasbenguedda.fr/projects/barriere_option.html

5.8 Features of Athena products

We have seen the basic Athena product, its construction, its sensitivities and its hedging. Now, the investor could want to have better guarantees or a higher coupon. For this purpose, we can add some interesting features to the product that we will expose in the following parts of this paper.

5.8.1 Exercise frequency

First thing first, the exercise frequency can be changed. Instead of looking at the level of the underlying stock, one could want to have a daily exercise frequency. This change of the frequency would have as an impact to increase the value of c^*, to remunerate the investor bigger risk taken.

If we change in our product the observation frequency from quarterly to daily, we have the following results:

		Base case	**Daily observation**
Contract	Number of observations	16	1008
	Observation frequency (years)	0.25	1/252
	First call date (years)	0.25	1/252
Pricing	Price (% nominal)	97.80	99.65
	95% confidence interval (±)	0.11	0.05
	Fair coupon (% p.a.)	10.54	11.74
Scenarios	Probability to reach maturity (%)	13.29	1.80
	Probability of a capital loss (%)	5.22	0.78
	Expected loss at maturity (% nominal)	3.05	0.46
	Expected life (years)	1.03	0.13
	Expected number of coupons	0.87	0.98
	Expected coupon paid (% nominal)	3.01	0.38

Table 6: Comparison between quarterly and daily autocall observation, all other parameters being those of the base case.

We observe that the fair coupon is now 11.74%.

5.8.2 Step-down barrier

The step down feature is really easy to understand. In fact, periodically, the auto call barrier is moving. It can be in both side. Generally the barrier is decreasing over time. We took the example of a 1% decreasing barrier at each observation date. That means at the end of the life of the product (if we don't consider the maturity as a redemption date), the barrier will be at 85% of the initial spot price. We have the following result:

		Base case	**Step-down 1%**
Contract	Last autocall barrier (%)	100	85
	Step-down per observation (%)	0	1
Pricing	Price (% nominal)	97.80	98.81
	95% confidence interval (±)	0.11	0.10
	Fair coupon (% p.a.)	10.54	8.24
Scenarios	Probability to reach maturity (%)	13.29	8.12
	Probability of a capital loss (%)	5.22	4.44
	Expected loss at maturity (% nominal)	3.05	2.62
	Expected life (years)	1.03	0.87
	Expected number of coupons	0.87	0.92
	Expected coupon paid (% nominal)	3.01	3.29

Table 7: Effect of a step-down autocall barrier, decreasing by 1% per quarterly observation from 100% to 85%, all other parameters being those of the base case.

5.8.3 Airbag

The airbag effect is also easy to understand

5.8.4 Buffer

5.8.5 Guaranteed capital

5.8.6 Coupon range accrual

5.8.7 Duo mechanism

5.8.8 Look back

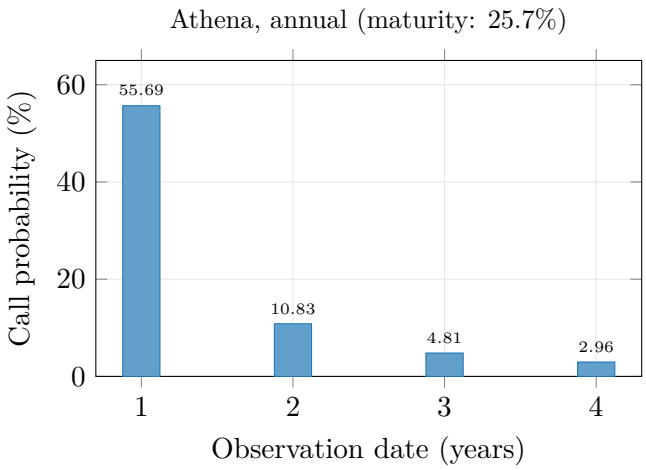


Figure 6: Base Athena: probability of being called at each observation date (autocall barrier at 100%).

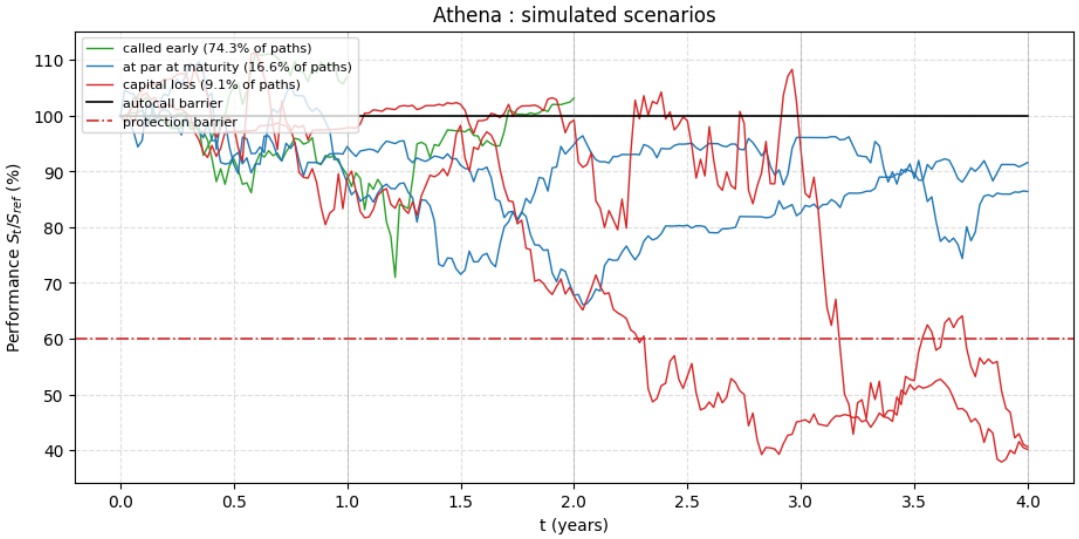


Figure 7: Base Athena: simulated paths of the performance, grouped by outcome.

5.9 All the forms of the Athena

Table **??** changes one feature of the base case at a time. All prices are computed at the same coupon of 8% (standard errors between 0.03 and 0.08); the fair coupon is the comparable quantity.

Reading the table. The investor is short a down-and-in put whose premium is paid back through the coupons; the fair coupon per annum is roughly this premium divided by the expected life of the note.

- *Observation frequency.* More frequent observations call the note earlier (expected life from 2.06 to 0.59 years) and reduce the probability of a loss (from 9.06% to 3.13%), but the put premium is spread over a much shorter life: the fair coupon rises from 9.41% to 11.11%.

Variant	Price at 8%	Fair coupon (% p.a.)	P(call t_1) (%)	P(maturity) (%)	P(loss) (%)	E[loss] (%)	E[life] (y)
Base (annual, snowball, AC 100%, EU 60%)	98.60	9.41	55.69	25.70	9.06	5.27	2.06
Semi-annual observations	98.64	9.93	53.30	18.61	6.97	4.08	1.48
Quarterly observations	98.77	10.54	55.43	13.29	5.22	3.05	1.03
Monthly observations	99.16	11.11	56.21	7.77	3.13	1.83	0.59
Fixed coupon (no snowball)	96.40	13.00	55.69	25.70	9.06	5.27	2.06
Autocall barrier 90%	100.95	7.03	77.37	13.24	6.41	3.77	1.54
Autocall barrier 110%	95.40	12.71	25.07	45.60	11.77	6.77	2.85
Step-down 5% per year	100.65	7.46	55.69	17.56	8.57	5.01	1.95
Quarterly, non-call 1 year	99.09	8.94	0.00	22.06	8.42	4.92	1.89
Protection 50%	99.90	8.10	55.69	25.70	5.91	3.86	2.06
Protection 70%	97.33	10.68	55.69	25.70	12.96	6.63	2.06
American protection (continuous)	96.83	11.18	55.69	25.70	16.43	7.20	2.06
American protection (daily)	96.95	11.05	55.53	25.67	16.07	7.18	2.07
Airbag	100.89	7.10	55.69	25.70	9.06	2.75	2.06
Capital guaranteed 90%	102.56	5.42	55.69	25.70	9.06	0.91	2.06
Capital guaranteed 100%	103.39	4.59	55.69	25.70	0.00	0.00	2.06
Buffer							

Table 8: Athena variants on the CAC40 (Heston, 50,000 paths). E[loss] in % of the nominal.

- *Snowball vs fixed coupon.* A fixed coupon pays c once whatever the call date, instead of $c\,t_k$: the fair coupon is 13.00% instead of 9.41%.
- *Autocall barrier.* A barrier at 90% calls 77% of the notes after one year and lowers the fair coupon to 7.03%; at 110%, 46% of the notes reach maturity and the fair coupon is 12.71%. A step-down of 5% a year lowers it to 7.46%.
- *Protection.* Each 5 points of protection barrier move the fair coupon by 0.4 to 0.7 point (Figure **??**). The American protection raises the probability of a loss from 9.06% to 16.43% and the fair coupon by 1.77 points; a daily monitoring lowers the fair coupon by 0.13 point with respect to a continuous monitoring.
- *Airbag and guarantee.* The airbag halves the expected loss (2.75% instead of 5.27%) for a coupon of 7.10%; a 100% capital guarantee leaves a coupon of 4.59%, the value of the call-at-par feature on top of a zero-coupon bond.

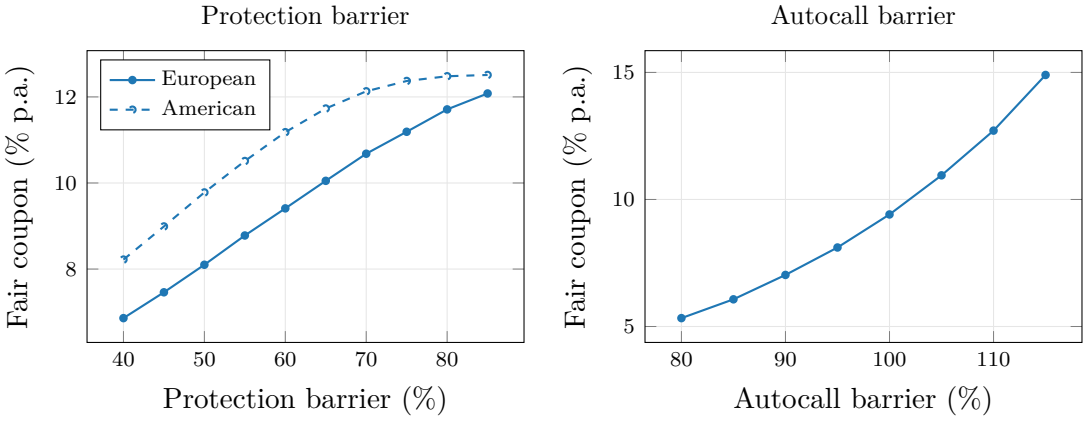


Figure 8: Athena: fair coupon as a function of the barriers (other features of the base case unchanged; American: continuous monitoring).

5.10 Greeks

Table **??** gives the Greeks of five Athena variants. Gamma and Volga are not significant at this number of paths (base case: Gamma $-3.0\times10^{-6}\pm5.4\times10^{-6}$, Volga -0.042 ± 0.037); the other

Greeks have relative standard errors between 3% and 45% (base case: Delta 0.00506 ± 0.00019, Vega -0.549 ± 0.016, Rho -0.77 ± 0.07, Dividend -1.12 ± 0.07, Theta 0.0050 ± 0.0016).

Athena	**Price**	**Delta** (per point)	**Delta 1%** (per 1% move)	**Vega** (per vol pt)	**Vanna** ($\times 10^4$)	**Rho** (per 1%)	**Dividend** (per 1%)	**Theta** (per day)
Base	98.60	0.00506	0.407	−0.549	2.51	−0.770	−1.115	0.0050
American protection	96.83	0.00644	0.518	−0.657	1.97	−0.546	−1.269	0.0050
Step-down 5%	100.65	0.00321	0.258	−0.637	6.36	−0.900	−0.935	0.0079
Airbag	100.89	0.00343	0.276	−0.354	1.28	−1.161	−0.815	0.0037
Guaranteed 90%	102.56	0.00241	0.193	−0.139	0.44	−1.436	−0.608	0.0023

Table 9: Greeks of Athena variants (nominal 100, coupon 8%).

The note is long the index (a 1% rise is worth 0.41 on 100) and short volatility: one vol point costs 0.55, the short put dominating the long digital-like call features. It is also short dividends: one point of carry lowers every forward and costs 1.12, twice the vega. It loses with higher rates (discounting of the coupons and of the nominal) and earns about 0.005 a day (1.8 a year) as the put decays. The capital guarantee reduces every risk but the rate risk, since the guaranteed redemption behaves as a zero-coupon bond. Figure **??** displays the table with 95% confidence intervals and Figure **??** the Greeks as functions of the spot, the initial fixing staying at 8042: the Delta increases as the spot falls towards the protection barrier (4825), and peaks just above it with an American protection.

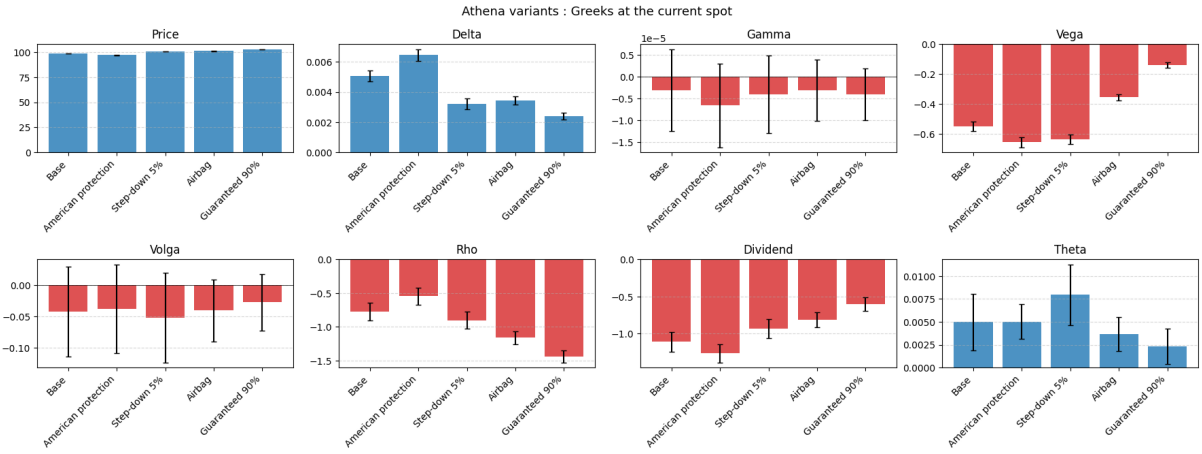


Figure 9: Greeks of the Athena variants with 95% confidence intervals.

Sensitivities to the Heston parameters. Per unit of parameter, the base Athena has $\partial V/\partial v_0 = -69.9$, $\partial V/\partial \theta = -69.4$, $\partial V/\partial \kappa = -0.81$, $\partial V/\partial \xi = 3.94$ and $\partial V/\partial \rho = -1.03$. The note loses with the variance level (short put) and gains with the vol-of-vol ξ.



Figure 10: Athena: price and Greeks as functions of the spot (30,000 paths, spot step 400).

6 Implementation

6.1 Code

- `Products/Athena/athena.py`: the self-contained class `Athena` (contract, `path_payoff`, `fair_coupon`, `statistics`);
- `Equity_Models/monte_carlo.py`: `MonteCarloPricer`, which simulates the paths and calls `path_payoff` for products with several cash flows;
- `Equity_Models/Heston.py`, `Equity_Models/forward_curve.py`: calibrated model and forward curve;
- `Equity_Models/greeks.py`: Greeks with common random numbers, including the Dividend Greek;
- `Products/market.py`: CAC40 market data and calibrated Heston model (`cac40_heston`);
- `Products/notes_analysis.py`: comparison of variants and plots;
- notebooks `Products/Athena/athena_pricer.ipynb` and `Products/Athena/athena_synthetic_pricing.ipynb`.

```
from market import cac40_heston
from athena import Athena
from Equity_Models.monte_carlo import MonteCarloPricer
from Equity_Models.greeks import Greeks

mkt = cac40_heston()
pricer = MonteCarloPricer(n_paths=50_000, steps_per_year=52, seed=42)
note = Athena(4.0, mkt.S0, 0.08, coupon_type="snowball", frequency=1.0,
              autocall_barrier=1.0, protection_barrier=0.6, protection_type="
                  european")
print(pricer.price(mkt.heston, note))              # price and standard error
print(note.fair_coupon(pricer, mkt.heston))        # coupon p.a. at par
print(note.statistics(pricer, mkt.heston))         # call probabilities, loss,
    life, coupon
print(Greeks(pricer).compute(mkt.heston, note, Greeks.ORDER))
```

6.2 Cost

With N paths and M steps, the simulation costs $\mathcal{O}(NM)$ in time and stores S and v, i.e. $16(M+1)N$ bytes: 167 MB for $N = 50{,}000$ and the weekly grid of 4 years, 807 MB on a daily grid. Following the note through its n observation dates costs $\mathcal{O}(nN)$, the American continuous protection $\mathcal{O}(NM)$. Measured times (one CPU core): 0.30 s for a valuation including the simulation; 0.02 to 0.12 s for the price, the fair coupon and the statistics of a variant on cached paths (1.5 s with an American protection); 26 s for the eight Greeks of five notes (14 valuations each); 41 to 44 s for the spot profiles of four notes (16 spots $\times$ 6 scenarios, 30,000 paths).

7 Limitations and Next Steps

- *Single underlying.* Most autocalls sold today are worst-of on two to four indices or stocks; they need a multi-asset model and a correlation calibration.
- *Model.* The value of an autocall depends on the forward skew and on the joint dynamics of spot and volatility between observation dates; a pure Heston model with constant parameters is not the market standard, which is a local stochastic volatility model calibrated to the whole surface [**?**].
- *Fair coupon.* It prices the note at par without issuer margin, funding spread or issuer credit risk; a real issue price is below par by the margin.
- *Discretisation.* Euler full truncation with the Feller condition violated ($2\kappa\theta/\xi^2 = 0.49$), weekly grid with the variance frozen over each week for the Brownian bridge.
- *Greeks.* Gamma and Volga are not significant with 50,000 paths: the payoff jumps at the autocall and protection barriers.
- *Dividends.* Continuous dividend yield bootstrapped from the forwards; real notes are exposed to discrete dividends and to dividend risk, which the Dividend Greek only measures for a parallel shift.

8 Conclusion

A single parametrisation covers the Athena with all its usual features, and a Monte Carlo pricer under a calibrated Heston model recovers three exact limit cases within one standard error. On the CAC40 of 12 February 2025, the base 4-year Athena pays 9.41% a year; the protection type is the most expensive feature for the investor (1.77 point of coupon for an American protection); more frequent observations raise the fair coupon, and a step-down of the autocall barrier lowers it. The note is short volatility and short dividends, the dividend exposure being twice the vega.

A Notation

Symbol	Meaning
N, S_{ref}, T	Nominal, initial fixing, maturity
t_k, $\delta_k = t_k - t_{k-1}$	Observation dates and accrual periods
$P_k = S_{t_k}/S_{\text{ref}}$	Performance at t_k
B_k^{call}, B^{prot}	Autocall and protection barriers (fractions of S_{ref})
c, $c^\star$	Coupon rate per annum, fair coupon
K_{put}, ϕ	Strike of the short put, capital floor
τ	Call date
$\mathbf{1}_{\text{KI}}$	Knock-in event of the protection

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